

Kamran Heydarov

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SUMMARY

Data-driven and analytical aspiring Data Scientist with a strong academic foundation and hands-on experience in statistical data analysis, machine learning, and thermodynamic modeling. Experienced in chemical analysis, chromatographic and spectroscopic techniques, and thermophysical research. Successfully developed and deployed open-source Python libraries for data processing and thermodynamic property estimation, demonstrating expertise in data engineering and scientific computing. Passionate about integrating advanced data analytics and machine learning techniques to solve real-world problems and drive process efficiency and sustainable innovation.

EDUCATION

Baku, Azerbaijan

Graduation Date: Jun 2027

University of Strasbourg

Master of Science (M.Sc.) in Physical Chemistry/ Chemical Engineering

Baku, Azerbaijan

Graduation Date: Jun 2025

Azerbaijan State Oil and Industry University

Bachelor of Science (B.Sc.) in Chemical Engineering (Chemical Technologies Faculty)

WORK EXPERIENCE

Baku, Azerbaijan

Sep 2025 - Present

Azerbaijan State Oil and Industry University

Django developer & Chief Specialist

- Engineered Django applications and managed database schemas (e.g., MongoDB) to handle critical university data.
- Processed and structured data using Python for reporting, ensuring high data integrity.
- Led development initiatives using Agile methodologies to deliver timely platform upgrades.

Baku, Azerbaijan

Sep 2023 - Feb 2025

Azerbaijan State University of Oil and Industry

Data Oriented Lab Assistant Supervised by Dr. Leyla Vazirova ([email](#))

- Assisted professors in undergraduate labs at UFAZ and ASOIU.
- Strengthened love for teaching and research.
- Guided 20+ students per lab session, improving their understanding of experimental concepts and helped boost lab grades.
- Taught students statistical methods for chemical analysis, enhancing their data interpretation skills and analytical reasoning.

PROJECT EXPERIENCE

Baku, Azerbaijan

March 2025 - Present

pyionics python library ([link](#))

owner & developer

- Developed a Python-based tool for thermodynamic property estimation of ionic liquids and mixtures.
- Integrated models for phase equilibrium and heat capacity prediction to support research and education.
- Optimized the codebase for usability in academic and industrial thermodynamic analysis.

Baku, Azerbaijan

Feb 2025 - Present

ChemGraph - Project Simulation Software

Executive Director

- Lead the development of ChemGraph, an open-source PFD design simulator supported by Azerbaijan State Oil and Industry University.
- Oversee its application in education and research, aiding students and researchers in process simulation.

- Enhance the platform’s functionality to improve accessibility and usability for academic and industrial applications.

Baku, Azerbaijan
Sep 2024 - Present

ILThermoGUI

Data Scientist (Group of Shamkhal Baybakev) ([email](#))

- Developed a Python library for installing and accessing NIST ILThermo data.
- Cleaned, processed, and structured datasets for improved usability.
- Designed visualizations to analyze and interpret thermophysical data.
- Conducted statistical analysis to extract meaningful insights.
- Researched ionic liquids, focusing on their thermophysical properties and applications.

Baku, Azerbaijan
Jan 2024 - Present

MentorCorp

Co-Founder & CTO

- Developed a user-friendly interface on the mentoring platform. Streamlined resource access.
- Achieved an average onboarding time reduction of 30% for new users seeking guidance in professional growth.

Baku, Azerbaijan
Jan 2024 - Sep 2024

YOUNGH

Founder & CEO

- Built a Q&A platform with 200+ active users, enabling students to earn up to \$500 collectively in its first 6 months.
- Enabled event tracking and provided opportunities through the website.
- Sold the platform to a company in our competition, expanding its reach and impact.

SKILLS & INTERESTS

Advanced Knowledge: Python (matplotlib, pandas, NumPy, tkinter, rdkit), LATEX, KNIME, Excel, MongoDB, Density Function Theory for calculation molecular quantum dynamics, Mathlab, Solidworks

INCUBATION PROGRAMS

Baku, Azerbaijan
Jan 2024 - May 2024

SABAH.HUB Incubation

Project Owner - Youngh

- Built a Q&A platform with 200+ active users, enabling students to earn up to \$500 collectively in its first 6 months.
- Enabled event tracking and provided opportunities through the website.

Baku, Azerbaijan
Oct 2023 - July 2024

UFAZ - IGNITE Incubation Program

Project Owner - Youngh

- Offers a comprehensive platform for academic and professional growth.
- Sold the platform to a company in our competition, expanding its reach and impact

CERTIFICATIONS

Data Manipulation with pandas | ELA982 SChE® Managing Lab Process Safety 1 | ELA986 SChE® Managing Lab Process Safety 2 | ELA954 SChE® Certificate Program - Lab Safety | MongoDB Python Developer Path | Excel Fundamentals | GitHub Foundations | KNIME Fundamentals | JavaScript (intermediate) Certificate | intermediate Python

Languages: Azerbaijani: Native, English: Fluent, Turkish: Native, Russian: B1